

Bimukti Mozzumdar

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U.S. Permanent Resident (Green Card) · No sponsorship required

EDUCATION

CUNY Hunter College

Bachelor of Arts in Computer Science, Minor in Mathematics; GPA: 3.93/4.0

- **Honors:** Thomas Hunter Honors Program, Dean's List

New York, NY

Expected Jan. 2028

EXPERIENCE

Machine Learning Research Assistant

Mar. 2026 – Present

CUNY Hunter College

New York, NY

- Developing a two-stage diffusion model for synthetic RNA-seq generation to model 20,000+ high-dimensional genomic features using a VAE with Gumbel-Softmax and an MLP denoiser.
- Enhanced style consistency for game art generation by 16% as measured by CLIP-similarity, fine-tuning SDXL with LoRA via pivotal tuning ([Kaggle](#)).

Software Engineering (Google-SWEP) Fellow

Mar. 2026 – Present

Google

New York, NY

- Selected as 1 of 250 from 1,900+ applicants for a year-long technical development program with 1:1 mentorship from a Google software engineer.
- Meet weekly for data structures, algorithms, and system-design coaching plus behavioral interview prep, reviewing solutions and industry engineering practices.

Software Engineering Intern

Jun. 2025 – Aug. 2025

Unadat

New York, NY

- Built an ML-powered alert system on NYC public school data to help case managers respond to at-risk students.
- Developed a Python pipeline to parse PDFs and emails into structured records, reducing manual data entry.
- Integrated the DocuSign API with a Node.js backend to automate authentication and contract signings.

Student IT Assistant

Oct. 2024 – Present

Social Sciences Computing Lab, CUNY Hunter College

New York, NY

- Support 300+ students weekly across 43 lab machines, handling OS imaging, hardware/software troubleshooting, account provisioning, and research-software upkeep (R/RStudio, ArcGIS Pro, NVivo).

PROJECTS

Redis Clone | *Go, TCP, Hash Tables, Concurrency* | [GitHub](#)

Apr. 2026 – Present

- Built an in-memory key-value store in Go with a custom hash table (MurmurHash3, separate chaining) and progressive rehashing to avoid O(n) latency spikes on resize.
- Designed a typed, length-prefixed binary wire protocol and a goroutine-per-connection TCP server over a mutex-guarded store; added TTLs with lazy (on-access) and active (bounded-sweep) expiration.
- Verified concurrency with Go's race detector under a multi-goroutine stress test and benchmarked the structure against sync.Map and map+RWMutex; enforced quality with GitHub Actions CI.

DegreeFlow | *React, TypeScript, Node.js, Prisma, PostgreSQL* | [GitHub](#) | [Live Demo](#)

May 2026 – Present

- Led the backend of an automated degree-audit tool, building a scheduler that generates prerequisite-valid course plans from a student's remaining requirements across an 8,800+ course catalog.
- Built a course recommender that pairs each suggested class with its top-ranked instructor, scoring professors by a Bayesian-weighted rating over scraped RateMyProfessors data to optimize for teaching quality.

PeerDrop | *Go, WebRTC, JavaScript, Docker* | [GitHub](#)

Oct. 2025

- Developed a browser-based P2P file-sharing tool with WebRTC and a Go signaling server for file transfers with no centralized storage backend, deployed across 43 lab workstations.

TECHNICAL SKILLS

Languages: Go, Python, C++, TypeScript, JavaScript, SQL

Frameworks: React, Node.js, Express, Hugging Face Diffusers, Django

Developer Tools: Git, Docker, Linux, AWS S3, PostgreSQL, Redis, Prisma, OpenAI API

Libraries: Pandas, NumPy, Scikit-learn, Puppeteer, PyTorch